

DTSA 5743

Cryptography and Information Theory

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (41.7 / 100)	34.8 / 100
SPECIALIZATION Applied Cryptography		

35

CURRENT AVG SCORE
across 6 job roles

75

POTENTIAL SCORE
with recommended additions

+58

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course combines cryptography (the techniques for protecting information from unauthorized access) and information theory (the study of information coding and transfer). More specifically, the course studies cryptography from the information-theoretical perspectives and discusses the concepts such as information entropy and the adversary’s knowledge capability, e.g., Kerckhoff’s Principle. It also contrasts the information-theoretic security and computational security to highlight the different train of thoughts that drive the cryptographic algorithmic construction and security analyses.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	41.7	74.5	+32.8	HIGH
ML Engineer	35.8	66.5	+30.8	HIGH
Data Scientist	35.0	88.1	+53.1	HIGH
Applied Scientist	34.1	92.5	+58.4	HIGH
Machine Learning Engineer	33.7	72.9	+39.1	HIGH
MLOps Engineer	28.5	53.6	+25.1	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

3 of 6 roles

systems

Business Communication

1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2

LOW

Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

41.7 → 74.5 (+32.8 pts)

data models learning build software technical code systems

ML Engineer

35.8 → 66.5 (+30.8 pts)

media discovery data learning machine users systems models

Data Scientist

35.0 → 88.1 (+53.1 pts)

data product analytics social build people problems approaches

Applied Scientist

34.1 → 92.5 (+58.4 pts)

discovery scientific research literature interdisciplinary problems models technical

Machine Learning Engineer

33.7 → 72.9 (+39.1 pts)

data learning analytics tools solutions protection business mobile

MLOps Engineer

28.5 → 53.6 (+25.1 pts)

data systems learning machine media tools models development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.